

2 Chemistry content

1	Principles of chemistry	11
2	Inorganic chemistry	16
3	Physical chemistry	21
4	Organic chemistry	23

1 Principles of chemistry

The following sub-topics are covered in this section.

- (a) States of matter
- (b) Elements, compounds and mixtures
- (c) Atomic structure
- (d) The Periodic Table
- (e) Chemical formulae, equations and calculations
- (f) Ionic bonding
- (g) Covalent bonding
- (h) Metallic bonding
- (i) Electrolysis

(a) States of matter	
Students should:	
1.1	understand the three states of matter in terms of the arrangement, movement and energy of the particles
1.2	understand the interconversions between the three states of matter in terms of: • the names of the interconversions • how they are achieved • the changes in arrangement, movement and energy of the particles.
1.3	understand how the results of experiments involving the dilution of coloured solutions and diffusion of gases can be explained
1.4	know what is meant by the terms: • solvent • solute • solution • saturated solution.
1.5C	know what is meant by the term solubility in the units g per 100 g of solvent
1.6C	understand how to plot and interpret solubility curves
1.7C	<i>practical: investigate the solubility of a solid in water at a specific temperature</i>